Extrema and Graphs of Functions

MATH 145 – Applied Calculus

Torin Quinlivan

October 25, 2024

$f'(x)$, $f''(x)$ and the graph of $f(x)$

Suppose we want to sketch the graph of $f(x) = x^2 + x - 12$.

We could plug in numbers for x and then plot those points until we can draw the graph.

However, we can use what we know about derivatives to make it easier.

$f'(x)$, $f''(x)$ and the graph of $f(x)$

Recall that:

- If $f'(x) > 0$, then $f(x)$ is increasing.
- If $f'(x) < 0$, then $f(x)$ is decreasing.
- If $f''(x) > 0$, then $f(x)$ is concave up.
- If $f''(x) < 0$, then $f(x)$ is concave down.

What happens when $f'(x) = 0$ or $f''(x) = 0$?

Critical Points

Critical Point

If $f(x)$ is a function, a **critical point** of $f(x)$ is a value p such that $f'(p) = 0$, or undefined. The **critical value** is the value of $f(p)$.

A *local minimum* or a *local maximum* of $f(x)$ *may* exist at a critical point of $f(x)$, so we can identify potential *local extrema* of $f(x)$ by solving the equation $f'(x) = 0$.

The First Derivative Test for Local Extrema

The First Derivative Test

Given a critical point p of a function $f(x)$, as you move left to right:

- if $f'(x)$ switches from negative to positive, then $f(x)$ has a local minimum there.
- if $f'(x)$ switches from positive to negative, then $f(x)$ has a local maximum there.

Example: The First Derivative Test

Example: Find all critical points of the function $f(x) = x^3 - 6x^2 + 9x$. Sketch a graph.

Example: Find all critical points of the function $f(x) = x^4 - 8x^2$. Sketch a graph.

The Second Derivative Test for Local Extrema

The Second Derivative Test

Given a critical point p of a function $f(x)$:

- if $f''(p) > 0$, then $f(x)$ has a local minimum at p.
- if $f''(p) < 0$, then $f(x)$ has a local maximum at p.
- if $f''(p) = 0$, then $f(x)$ could do anything at p.

Example: The First Derivative Test

Example: Find all local minima and maxima of the function $f(x) = x^3 - 6x^2 + 9x$. Sketch a graph.

Example: Find all local minima and maxima of the function $f(x) = x^4 - 8x^2$. Sketch a graph.

Concavity and the Second Derivative

Inflection Point

A point p at which the *concavity* of $f(x)$ changes is called an **inflection point**.

Recall that:

- If $f''(x) > 0$, then $f(x)$ is concave up.
- If $f''(x) < 0$, then $f(x)$ is concave down.

So to find inflection points, solve $f''(x) = 0$ and check $f''(x)$ on either side of the inflection point to see if it changes sign.

Example: The First Derivative Test

Example: Find all inflection points of the function $f(x) = x^3 - 6x^2 + 9x$. Sketch a graph.

Example: Find all inflection points of the function $f(x) = x^4 - 8x^2$. Sketch a graph.

Example: Hard Calculus and Algebra

Example: Sketch a graph of $f(x) = x^2 \ln(x)$.